

ALDDREU AF, Korea, Republic of

WMO: 471870

Lat: 33.200N

Lon: 126.267E

Elev: 27

StdP: 101.00

Time Zone: 9.00 (E09)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.3	-0.6	-6.9	2.1	1.0	-5.9	2.3	1.3	18.3	1.6	17.4	2.0	9.2	350	0.504

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.1	31.0	26.4	30.0	26.2	29.0	25.6	27.1	29.6	26.6	28.9	26.1	28.2	4.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.7	28.3	26.0	21.5	28.1	25.5	20.8	27.6	85.5	30.0	83.5	29.2	81.6	28.0	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.9	14.4	12.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			-4.8	28.2	2.0	0.8	-6.3	28.8	-7.4	29.3	-8.5	29.8	-10.0	30.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.4	4.9	5.2	8.3	12.8	17.2	21.0	25.3	26.6	23.0	18.3	13.4	7.7
	DBStd	7.90	3.14	3.40	3.10	2.89	2.21	1.83	2.16	1.67	2.35	2.72	3.53	3.46
	HDD10.0	474	159	138	69	9	0	0	0	0	0	0	11	88
	HDD18.3	1823	415	367	311	165	48	2	0	0	1	35	151	329
	CDD10.0	2439	2	4	17	94	224	331	475	516	389	257	112	17
	CDD18.3	746	0	0	0	0	13	83	217	257	140	33	2	0
	CDH23.3	4766	0	0	0	0	15	170	1562	2301	660	58	1	0
	CDH26.7	1029	0	0	0	0	0	8	341	574	106	1	0	0
Wind	WSAvg	5.3	6.2	6.6	6.1	5.3	5.1	4.4	4.1	4.7	4.3	4.7	5.8	6.3
Precipitation	PrecAvg	1521	56	64	99	122	141	194	216	260	168	71	79	50
	PrecMax	2381	130	155	174	284	306	395	709	590	554	259	274	122
	PrecMin	868	9	12	14	39	63	46	13	76	13	4	2	8
	PrecStd	326	29	33	40	59	68	83	151	136	120	53	62	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.0	15.9	17.1	21.2	25.1	27.8	32.8	32.8	30.6	26.1	22.3	18.1
		MCWB	12.9	13.7	13.1	16.6	18.6	21.3	26.5	27.0	25.7	21.6	18.3	15.5
	2%	DB	13.0	13.8	15.5	19.2	23.2	26.1	30.9	31.2	29.1	24.9	20.9	16.1
		MCWB	10.6	11.7	12.9	15.6	18.3	21.1	26.4	26.6	24.8	20.4	17.3	13.2
	5%	DB	11.2	12.2	14.2	18.2	22.1	25.1	29.8	30.2	27.9	23.9	19.9	14.2
		MCWB	8.8	9.8	11.5	15.3	17.9	21.0	26.1	26.3	24.1	19.7	16.8	11.3
	10%	DB	9.9	10.8	13.2	17.2	21.1	24.1	28.4	29.7	26.8	22.7	18.4	13.0
		MCWB	7.8	8.2	10.5	14.6	17.5	20.9	25.6	26.1	23.4	18.8	15.6	10.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.8	14.2	15.2	18.3	21.1	24.3	27.5	28.0	26.9	23.4	19.7	16.1
		MCDB	14.7	15.4	16.0	19.6	22.9	25.2	29.9	30.2	29.1	24.9	21.1	17.9
	2%	WB	11.2	12.3	13.9	17.1	20.2	23.4	26.9	27.2	26.0	21.8	18.6	13.6
		MCDB	12.6	13.4	15.2	18.8	21.8	24.4	29.3	29.8	27.9	23.9	20.1	15.8
	5%	WB	9.2	10.2	12.4	16.1	19.4	22.6	26.4	26.8	25.4	20.6	17.2	11.9
		MCDB	10.9	12.0	13.9	17.5	21.0	23.8	28.6	29.3	27.2	22.7	19.1	14.2
	10%	WB	7.8	8.3	11.0	15.2	18.4	21.8	26.0	26.4	24.3	19.5	15.9	10.4
		MCDB	9.8	10.3	13.0	16.8	20.2	23.4	27.9	28.6	26.1	22.0	18.2	12.8
Mean Daily Temperature Range	5% DB	MDBR	5.4	5.9	6.5	6.8	6.5	5.7	4.6	5.1	6.3	6.8	6.3	5.4
		MCDBR	7.5	7.7	7.7	7.7	7.9	7.3	6.4	6.0	7.4	8.0	7.5	7.6
		MCWBR	6.6	6.4	6.1	5.9	5.2	4.5	3.5	3.6	4.8	5.7	6.1	6.6
	5% WB	MCDBR	7.2	7.2	6.6	6.5	6.0	4.9	5.0	5.3	5.9	7.0	6.9	7.2
		MCWBR	7.0	6.6	6.1	5.8	4.8	3.7	3.2	3.4	4.2	5.7	6.5	6.7
Clear-Sky Solar Irradiance	taub	0.470	0.509	0.558	0.567	0.573	0.601	0.585	0.539	0.496	0.470	0.467	0.464	
	taud	1.940	1.879	1.779	1.815	1.855	1.877	1.967	2.103	2.161	2.096	2.022	1.959	
	Ebn at Noon	704	724	725	744	744	721	730	758	773	755	704	680	
	Edh at Noon	151	177	211	212	206	201	183	157	143	141	138	141	
All-Sky Solar Radiation	RadAvg	1.98	2.83	4.07	4.85	5.19	4.55	5.06	5.21	4.30	3.74	2.55	1.84	
	RadStd	0.24	0.37	0.33	0.41	0.57	0.56	0.71	0.52	0.40	0.37	0.29	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page